

Preventing Cumulative Layout Shift (CLS)

How width and height attributes create pre-parsed layout constraints

🕒 Browser Parsing Timeline

1. HTML Payload Arrives

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2. UA Pre-scanner

Milliseconds before CSS executes,
the browser reads the raw attributes:

Width = 800 Height = 400

3. Proportional Fraction Math

Browser internally calculates the aspect ratio:

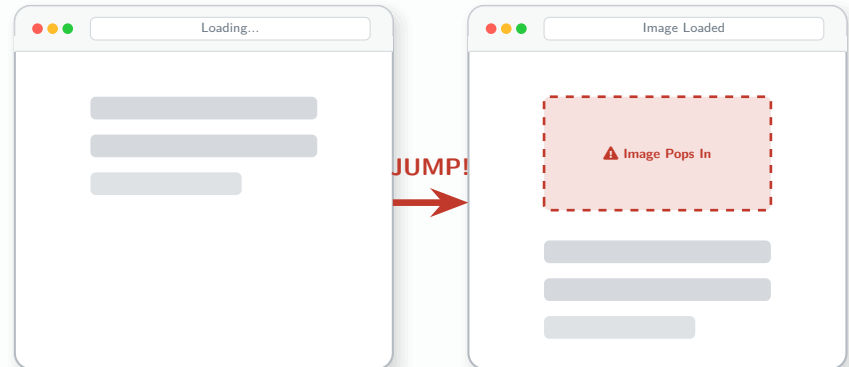
$800 \div 400 = 2 : 1$

4. Space Allocation

Layout engine creates an empty block maintaining the **2 : 1** ratio, reserving space *before* the image downloads.

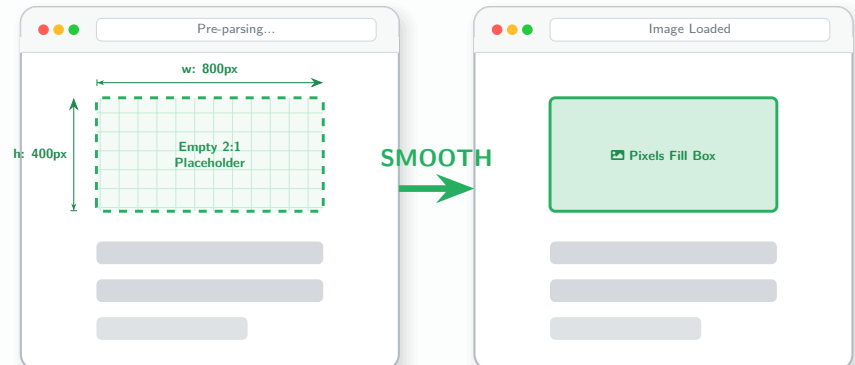
👁 User Experience (Visual Load)

❌ Scenario A: No Width/Height Attributes (CLS!)



Content Shifts Wildly.
Browser doesn't know image size until downloaded.

✅ Scenario B: With Attributes (Smooth Load)



Zero Layout Shift.
Pre-parser creates exact boundaries immediately.